
BPM2_NB_ESOFT QUIZ 8_S2026_NU

of Questions: 30

Question #: 1

A 68-year-old man comes to the physician because of a 2-year history of slowly worsening vision. He now finds himself squinting to read menus when he goes out to dinner and recently misread the directions on a medication bottle causing him to put ear drops into his eyes. Physical examination shows normal appearing eyes. Which of the following diagnoses is most likely in this patient?

- A. Hyperopia
- B. Astigmatism
- ✓C. Presbyopia
- D. Myopia
- E. Cataract

Rationale: Presbyopia is a type of refractive error that causes close objects to gradually appear out of focus as the eye ages. It is caused by loss of the elasticity of the lens inside the eye due to natural aging. Presbyopia eventually affects almost everyone, beginning as early as age 35, and progressing over the next 20 to 30 years. Hyperopia can affect anyone, regardless of age.

Item Description: ESoft.quiz.BEHS.0021.HD3_over 65_presbyopia

Question #: 2

A 40-year-old man comes to the physician because of chronic anxiety. He is unmarried and works full-time as a physician assistant. He constantly worries about his future and struggles to find compromises. He feels he is running out of time to reach his life goals. He has no history of psychiatric symptoms and denies negative childhood experiences. Which of the following stages of Erikson's psychosocial development most likely applies to this man?

- ✓A. Generativity vs stagnation

- B. Intimacy vs isolation
- C. Identity vs role confusion
- D. Industry vs inferiority
- E. Shame vs doubt

Rationale: During middle adulthood, individuals often face the conflict of generativity vs. stagnation. Generativity refers to the desire to contribute to society and leave a legacy, whether it be through raising children, mentoring others, or making a meaningful impact in their career. Stagnation, on the other hand, refers to a sense of feeling unproductive and lacking purpose. This conflict arises as individuals reflect on their accomplishments and question whether they have made a meaningful difference in the world.

Item Description: ESoft.quiz.BEHS.0010.HD3_Erikson stages_generativity vs stagnation

Question #: 3

A boy is born with a large swelling of the proximal end of his umbilical cord. Physical examination of the swelling shows coils of small bowel in a membranous sac. Which of the following best describes the baby's condition?

- ✓A. Omphalocele
- B. Umbilical hernia
- C. Gastroschisis
- D. Ileal diverticulum
- E. Omphaloenteric fistula

Rationale: This patient has an omphalocele which is a persistence of the physiological hernia of the midgut. This herniated gut is contained within the greater sac.

Item Description: A boy is born with a

Question #: 4

A 73-year-old man is evaluated for loss of general sensation on the left half his tongue anterior to the terminal sulcus following surgery to resect a tumor in the oral cavity near the posterior aspect of the body of the mandible. Which of the following best describes the development of this part of the tongue?

- ✓A. Lateral lingual swelling of the first arch
- B. Medial lingual swelling of the first arch
- C. Copula of the second arch
- D. Hypopharyngeal eminence of the third arch
- E. Hypopharyngeal eminence of the fourth arch

Rationale: The tongue develops from swellings from most of the pharyngeal arches. The lateral portion of the anterior 2/3 is formed from the lateral lingual swelling of the first arch.

Item Description: A 73-year-old man is

Question #: 5

A 21-year-old man comes to the physician because of excessive mouth dryness for the past 2-weeks. Physical examination shows white and thick nodules in the mucosa of cheek and floor of the oral cavity. Also parotid duct is blocked by the nodules. Which of the following structures, responsible for mouth dryness is most likely innervated by?

- ✓A. Otic ganglion
- B. Ciliary ganglion
- C. Pterygopalatine ganglion
- D. Sympathetic ganglion

Rationale: Preganglionic parasympathetic fibers for the parotid secretion pass through the glossopharyngeal nerve and the parasympathetic postganglionic cell bodies for the parotid gland are located in the Otic ganglion and are relayed to the parotid gland via the auriculotemporal branch of the mandibular nerve. These fibers are responsible for the secretion of saliva from the parotid gland.

Item Description: Otic ganglion

Question #: 6

A 3-year-old boy is brought to the physician by his father because of a 2-month history of behavioral changes since the death of his mother. He used to enjoy going to preschool but now cries and clings to his father when he drops him off in the morning. The physician explains that the child is adjusting his existing schema about the world -- that people he loves and trusts can disappear -- which is making it hard for him to trust that his father will return to the preschool to pick him up each day. Which of the following terms best describes this process of creating a new cognitive schema?

- A. Object permanence
- B. Conservation
- C. Compensation
- D. Assimilation
- ✓E. Accommodation

Rationale: If new information is not too discrepant from previously learned material, then it can be incorporated into an existing schema, which Piaget called "assimilation." However, if there is a significant discrepancy, Piaget labeled the creation of a new schema as "accommodation."

Assimilation incorporates new information or experiences into existing cognitive structures or schemas. Information is interpreted based on existing knowledge and understanding of the world. New experiences or concepts fit into existing mental frameworks without significantly being altered. Example: if this child's mother was ill and could not take care of him for a short period of time, he may have been able to incorporate this into his existing relationship schema of loved ones as generally available to meet his needs.

Accommodation is the modification of existing schemas to incorporate new information or experiences. When new information or experiences do not fit existing schemas, adjustments or modifications to the mental framework is done to accommodate the new information. The complete loss of his mother could not be incorporated into the child's existing relationship schema, so he had to accommodate this new information by forming a new schema about close relationships.

Item Description: ESoft.quiz.BEHS.0007.HD1_Piaget_cognitive_accomodation

Question #: 7

An 18-year-old man is brought to the emergency department 2 hours after a motor vehicular accident. He was speeding along the highway after drinking lots of alcoholic beverages and crashed into a guard rail. He says his mood is low after a recent break-up from a long-term relationship. He has a 5-year history of major depressive disorder and is currently on medication. Physical examination is unremarkable. His blood alcohol level is above the legal limit for driving. Which of the following health conditions is most likely a leading cause of death in this patient?

- A. Cancer
- B. Heart disease
- C. Lower respiratory illness
- ✓D. Suicide
- E. Liver disease

Rationale: Suicide is the second most common cause of death for adolescents in the U.S. and is the most likely adverse outcome for this 18-year-old patient who has a history of depression along with other risk factors for suicide (e.g., substance abuse, and recent breakup).

Item Description: ESoft.quiz.BEHS.0017.HD2_death causes_adolescents

Question #: 8

A 56-year-old man is brought to the emergency department by his wife because of a 1-day history of erratic behavior. He has a 30-year history of chronic alcohol abuse. He is diagnosed with severe cirrhosis leading to hepatic encephalopathy. Laboratory studies show hyperammonemia. Which of the following additional laboratory findings is most likely in this patient?

- A. Elevated glycine
- B. Elevated serum albumin
- ✓C. Elevated serum glutamine
- D. Elevated blood urea nitrogen (BUN)
- E. Elevated serum methylmalonate

Rationale: Hepatic encephalopathy is characterized by high blood ammonia levels. This results in the incorporation of ammonia to glutamate to form glutamine and increases circulating glutamine levels. The blood ammonia levels are increased due to impairment of the urea cycle due to liver disease.

Item Description: QUIZ-Hepatic encephalopathy

Question #: 9

A 2-month-old boy is brought to the emergency department by his parents because of a 1-day history of seizures. Laboratory studies show hyperammonemia and high blood levels of argininosuccinate, glutamine, and citrulline. Which of the following is the most likely diagnosis?

- A. N-acetylglutamate synthetase deficiency
- B. Ornithine transcarbamylase deficiency
- C. Argininosuccinate synthetase deficiency
- ✓D. Argininosuccinate lyase deficiency
- E. Carbamoyl phosphate synthase-I deficiency
- F. Arginase deficiency

Rationale: The vignette describes a patient with an inherited urea cycle defect resulting in hyperammonemia. Elevated levels of argininosuccinate indicate the deficiency of argininosuccinate lyase.

Note that all inherited urea cycle defects are characterized by hyperammonemia.

Patients with argininosuccinate synthetase deficiency have elevated citrulline levels and low levels of argininosuccinate.

CPS-I deficiency is characterized by hyperammonemia and low levels of all other urea cycle intermediates.

N-acetyl glutamate (NAG) is an activator of CPS-I. Low NAG levels (NAG synthetase deficiency) are characterized by hyperammonemia.

Arginase deficiency is characterized by elevated arginine levels.

Item Description: BCHM.MB.2002; Urea - ASL defi

Question #: 10

A 4-month-old girl is brought to the physician by her parents because of failure to thrive. Physical examination shows coarse facial features, bilateral hip dislocation and restricted joint mobility. Laboratory tests show a 10-fold increase in iduronate sulfatase and β -hexosaminidase activity in the blood. However, there is no activity of these enzymes in cultured fibroblasts. Laboratory studies show a defect in sorting enzymes to the lysosomes. Which of the following is the most likely diagnosis?

- ✓A. I-cell disease
- B. Tay-Sachs disease
- C. Hurler syndrome
- D. Hunter syndrome
- E. Fabry disease
- F. Niemann-Pick disease

Rationale: The vignette describes a child with I-cell disease. It is due to a defect in sorting enzymes to the lysosomes. These enzymes are secreted into the blood and there are elevated levels of the lysosomal enzymes in circulation. There is accumulation of complex lipids (glycolipids and sphingolipids) and glycosaminoglycans in the lysosomes.

Item Description: Lysosomal storage diseases

Question #: 11

A 4-year-old boy is brought to the physician by his mother because of a 6-month history of bedwetting. He has been toilet trained since the age of 3 but started bedwetting again after the birth of his sibling. Which of the following defense mechanisms best explains his resumption of bedwetting?

- A. Acting out
- B. Displacement
- ✓C. Regression
- D. Suppression
- E. Reaction formation

Rationale: Regression involves the unconscious removal from consciousness of some threatening or unpleasant memory or event, and refers to the tendency of the individual to return to an earlier, less stressful, stage of development (or time in life), during which the individual's needs were satisfied and which was relatively stress-free. This defense mechanism is seen in times of crisis, stress, illness and other life challenges.

Acting out is similar in that it involves the indirect expression of emotion but involves the expression of the emotion through conduct disturbance. Suppression involves the voluntary, temporary removal from consciousness of some unpleasant or threatening future event.

Item Description: Esoft.quiz.BEHS.0064.PT.psychotherapies_defense mechanisms_regression

Question #: 12

A 6-year-old boy is brought to the physician by his father because of a 1-week history of severe ear pain, fever, cough, and discharge from the left ear. He has a history of recurrent ear infections. His temperature is 40°C (104°F). An otoscopy of the left ear shows a perforation of the tympanic membrane, with a purulent discharge. Audiometry testing shows reduced hearing in the affected ear. The physician suspects a chronic otitis media due to a chronic tympanic membrane perforation. Which of the following regions of the ear is most likely affected based on the physician's suspicion?

- A. External
- B. Cochlear canal
- C. Vestibule
- ✓D. Middle
- E. Semicircular canals

Rationale: The tympanic cavity/Middle ear is an air-filled space of the middle ear connected to the nasopharynx by the auditory tube. Because of this, infections of the middle ear is common in children with upper respiratory infections. Middle ear infections can also occur in cases of tympanic membrane perforations, as in the case of our 6-year old patient. The middle ear also contains the auditory ossicles.

The tympanic membrane separates the external ear from the middle ear. Therefore any breaches in this membrane can lead to access of the middle ear.

Options B, C and E all point to structures within the inner ear.

Item Description: HCB.1360 - Middle ear

Question #: 13

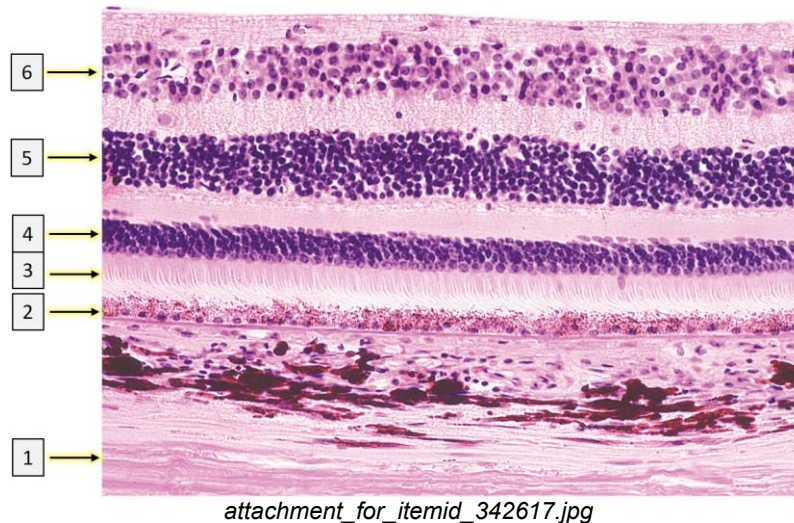
A 41-year-old man comes to the physician because of a 3-week history of visual changes in his right eye following an accidental electric current injury 4 weeks ago. A fundoscopic examination of the right eye shows blurring of the retinal vessels over areas of the retina, consistent with retinal detachment. Which of the following arrows in the attached photomicrograph best corresponds to the most likely area of injury leading to the examination findings?

- A. 1

- B. 2
- ✓C. 3
- D. 4
- E. 5
- F. 6

Rationale: Based on the description of the observations of the vascular findings on fundoscopy, this is classic for retinal detachment. Common site for retinal detachment is between the layers of pigmented epithelium (#2) and the layers of rods and cones(#3). Arrow 3 is closest to this region.

Attachment:



Item Description: HCB.1354 - Retinal detachment

Question #: 14

A 54-year-old woman comes to the physician because of a 1-week history of palpitations, and a 2-month history of lightheadedness and easy fatiguability. She underwent gastric bypass (bariatric) surgery 2 years ago. As a complication of the surgery, a large number of eosinophilic cells usually found at the neck of the gastric glands are lost. Which of the following vitamins is most likely decreased as a result of the loss of these cells?

- A. E
- B. A
- C. C
- ✓D. B12
- E. B9
- F. D
- G. K

Rationale: Parietal cells produce the HCL needed for the conversion of pepsinogen (produced by the

chief cells) into pepsin. They are eosinophilic cells located in the neck of the gastric glands. These cells also produce intrinsic factor, which is responsible for the absorption of vitamin B12 in the ileum of the small intestine. Deficiency of B12 results in Pernicious anemia. Since these cells were lost during the surgical procedure, there is less intrinsic factor, leading to less vitamin B12.

Item Description: HCB.1215 - Cells of gastric mucosa - Parietal

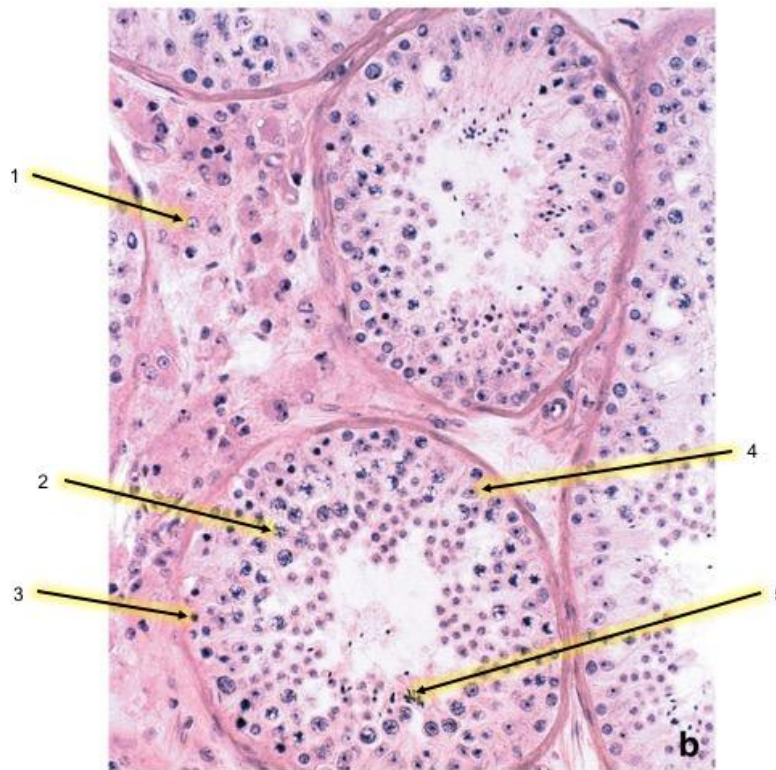
Question #: 15

A 7-year-old boy is brought to the physician by his mother because of a 5-month history of rapid growth and the development of pubic hair. Physical examination shows signs of precocious puberty including development of the external genitalia and the presence of pubic and facial hair. An ultrasound of his right testis shows a 3-cm intratesticular mass. Laboratory studies show elevated testosterone levels. Which of the following cells in the attached photomicrograph is most likely responsible for the production of the elevated hormone levels observed in this case?

- ✓A. 1
- B. 2
- C. 3
- D. 4
- E. 5

Rationale: Under the influence of Luteinizing hormone, Leydig cells or interstitial cells secrete testosterone. Leydig cell tumors are predominantly benign tumors that are hormonally active. Because of the elevated testosterone levels, this child will exhibit signs of puberty prematurely.

Attachment:



attachment_for_itemid_342614.jpg

Item Description: HCB.1091 - Function of Leydig cells

Question #: 16

A 58-year old man comes to the physician because of a 3-week history of shortness of breath, swollen feet, and chest discomfort. He has a 20-year history of chronic obstructive lung disease. His temperature is 37°C (98.6°F), pulse is 86/min, blood pressure is 126/83 mmHg, and respirations are 22/min. Physical examination shows jugular venous distention, pitting edema of the dorsum of feet, and tender hepatomegaly. ECG shows right axis deviation. Which of the following mechanisms best explains edema in this patient?

- A. Decreased oncotic pressure
- B. Impaired lymph drainage resulting in protein and fluid accumulation
- C. Increased pulmonary hydrostatic pressure secondary to left heart failure
- D. Decreased capillary hydrostatic pressure secondary to right ventricular failure
- ✓E. Increased central venous pressures secondary to right ventricular failure

Rationale: This patient has right heart failure because of lung disease. If the central venous pressure is increased it is reflected in the increase in jugular venous pressure and that can be estimated in an individual- visible pulsatile neck veins when the patient is semi recumbent.

Item Description: PHYS.0180. Heart failure Basis of edema

Question #: 17

A 67-year-old woman comes to the physician because of a 2-week history of back pain and weakness. She was diagnosed with breast cancer 6 months ago. Her vital signs are within normal limits. Laboratory studies show elevated plasma calcium levels. Radiographic imaging indicates lytic lesions in the body of the lumbar vertebrae and pelvis. Which of the following combination of plasma PTH and phosphate (PO₄) is most likely in this patient?

Option	Plasma PTH	Plasma PO ₄
A	Increase	No change
B	Increase	Decrease
C	Decrease	Increase
D	No change	Decrease
E	Decrease	Decrease

- A. A
- B. B
- ✓C. C
- D. D
- E. E

Rationale: C is the right answer. Breast cancer caused hypercalcemia. Elevated plasma calcium suppresses PTH secretion by negative feedback. In the absence of PTH, renal tubular reabsorption of PO₄ increases resulting in increased plasma PO₄.

Item Description: PHYS.1063. Effect of hypercalcemia on plasma PTH and PO₄

Question #: 18

A 76-year-old man comes to the physician because of a 1-month history of fecal incontinence. He says he is not able to hold his bowel and stool leaks from his rectum without warning. He does not experience urge to defecate. He has a 26-year history of type 2 diabetes mellitus. His vital signs are within normal limits. Physical examination shows no abnormalities. He is suspected of having defective functioning of the nerves supplying to the rectum and anal sphincters. Which of the following stimuli most likely initiates normal defecation reflex?

- A. Stimulation of the sympathetic nerves
- ✓B. Distension of the rectum
- C. Distension of the stomach
- D. Stimulation of the sacral anterior horn cells
- E. Stimulation of the lumbar anterior horn cells

Rationale: Normal defecation reflex begins when the rectum becomes full of stool. Distension of the rectum is the stimulus for initiating defecation reflex. The sensation of fullness of rectum reaches the

sacral segment of the spinal cord that controls defecation reflex. This signal relaxedly contracts the rectal wall and relaxes the internal and external anal sphincters and thus defecation occurs. This means that when the rectum is full, the defecation reflex will occur, emptying the bowel.

Item Description: PHYS.1116. Defecation reflex

Question #: 19

A 28-year-old woman comes to the physician for a routine examination. She is married for 2 years and wishes to have a child. Menses occur at regular 28-day intervals. She is not using any form of contraception. Her vital signs are within normal limits. Physical examination shows no abnormalities. Laboratory studies show normal levels of pituitary and ovarian hormones. She is currently on the 11th day of her menstrual cycle. Which of the following findings is most likely in her during this stage of the cycle?

- A. Low plasma estrogen
- B. Elevated basal body temperature
- ✓C. Thin cervical mucus
- D. High plasma progesterone
- E. Increased endometrial glandular secretions

Rationale: This woman is at the late follicular stage of her cycle. During this stage, there is elevated plasma estrogen being secreted from the matured graafian follicle. High estrogen makes the cervical mucus thin and highly elastic. The uterine endometrium is in a proliferative stage not secretory stage.

Item Description: PHYS.1080. Changes during the menstrual cycle

Question #: 20

A 24-year-old man comes to the physician because he requires follow-up care as part of a 1-year history of traumatic bilateral sectioning of the mid-thoracic spinal cord. Which of the following urinary patterns is most likely?

- A. Periods of nocturia
- ✓B. Reflexive
- C. Voluntary
- D. Continuous dribbling
- E. Retention

Rationale: The intactness of the sacral spinal cord and the associated peripheral nerves suggests reflexive urination. Retention, as part of spinal shock, would likely have been present for weeks or perhaps a few months after the injury.

Item Description: SOM.MK.I.BPM2.4.NB.2.NSCI.0280 Automatic and atonic bladder

Question #: 21

A 46-year-old man comes to the physician because of a 3-month history of unexplained loss of appetite and body weight. Physical examination shows a decrease of 6kg since his physical examination of last year. Which of the following nuclei is most likely expressing relative underactivity?

- A. Ventral lateral nucleus of the thalamus
- B. Dorsomedial nucleus of the thalamus
- C. Ventromedial nucleus of the hypothalamus
- ✓D. Lateral nucleus of the hypothalamus
- E. Periventricular nucleus of the hypothalamus
- F. Mammillary bodies

Rationale: Dysfunction of the lateral hypothalamic nuclei can suppress feeding behavior and promote weight loss. The effects are quite dramatic with advanced hypothalamic tumors which obliterate both medial and lateral hypothalami. In the absence of both lateral hypothalamic nuclei, feeding is not initiated, leading to emaciation.

Item Description: SOM.MK.1.BPM2.1.NB.1.NSCI.0281 Feeding body weight

Question #: 22

A 24-year-old man comes to the physician because of 2-hour history of restlessness and anxiety. He says that he has been anxious about an upcoming work-related presentation. His blood pressure is 150/92 mmHg, and his pulse is 100/min. He has cold clammy skin, and he sweats profusely. Which of the following structures is most likely overstimulating the HPA axis?

- A. Cingulate gyrus
- B. Frontal lobe
- C. Hippocampus
- D. Pineal
- ✓E. Amygdala

Rationale: Acute and chronic stress both lead to activation of amygdala which then stimulates the HPA axis to activate the sympathetic nervous system (quick and fast) and adrenal glands (both cortex and medulla) to release epinephrine, norepinephrine and cortisol. PNS is not activated in acute stress. The hippocampus and amygdala interact in relation to formation of stress-related memories.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0274 Amygdala and HPA axis

Question #: 23

An 46-year-old man comes to the physician because of a 4-day history of fever and emotional outbursts. His body temperature is 38.9 C (102.0 F). Laboratory studies are negative for pathogenic infection. Imaging studies show a tumor of the brain. Which of the following structures is most likely lesioned, leading to thermic dysregulation?

- A. Thalamus
- B. Epithalamus
- C. Subthalamus
- ✓D. Hypothalamus
- E. Amygdala
- F. Piriform cortex

Rationale: The hypothalamus acts as a comparator and often also as a sensor and an effector in relation to homeostatic regulation of vital functions such as body temperature.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0275 Hypothalamus central control unit

Question #: 24

A 27-year-old man comes to the physician because of a 4-month history of fatigue and decreased libido. Neurological examination shows a bitemporal hemianopia. Which of the following hormones is most likely failing to reach the anterior pituitary, thereby explaining the loss of libido?

- ✓A. Gonadotropin-releasing hormone
- B. Testosterone
- C. Luteinizing hormone
- D. Follicle stimulating hormone
- E. Epinephrine
- F. Antidiuretic hormone

Rationale: GnRH is released into the hypophyseal portal system for action on cells of the anterior pituitary, which release gonadotropins, including luteinizing hormone, which promotes the synthesis of testosterone (linked to a libido).

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0278 Hypothalamus-pituitary

Question #: 25

A 46-year-old man comes to the physician because of a 12-hour history of weakness of the legs. A nerve conduction velocity study shows that the weakness

is probably unrelated to demyelination. Which of the following motor nerve conduction velocities is most likely observed in this patient?

- A. .055 - .06 m/s
- ✓B. 55 - 60 m/s
- C. 550 - 600 m/s
- D. 5500 - 6000 m/s
- E. 55000 - 60000 m/s

Rationale: Conduction velocities of motor or sensory nerves are normally about 55 - 60 m/s. In peripheral demyelinating disease values below these levels are often recorded. Normal nerve conduction velocities often normalize, during recovery from transient peripheral demyelination.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0181 Nerve conduction velocity principles

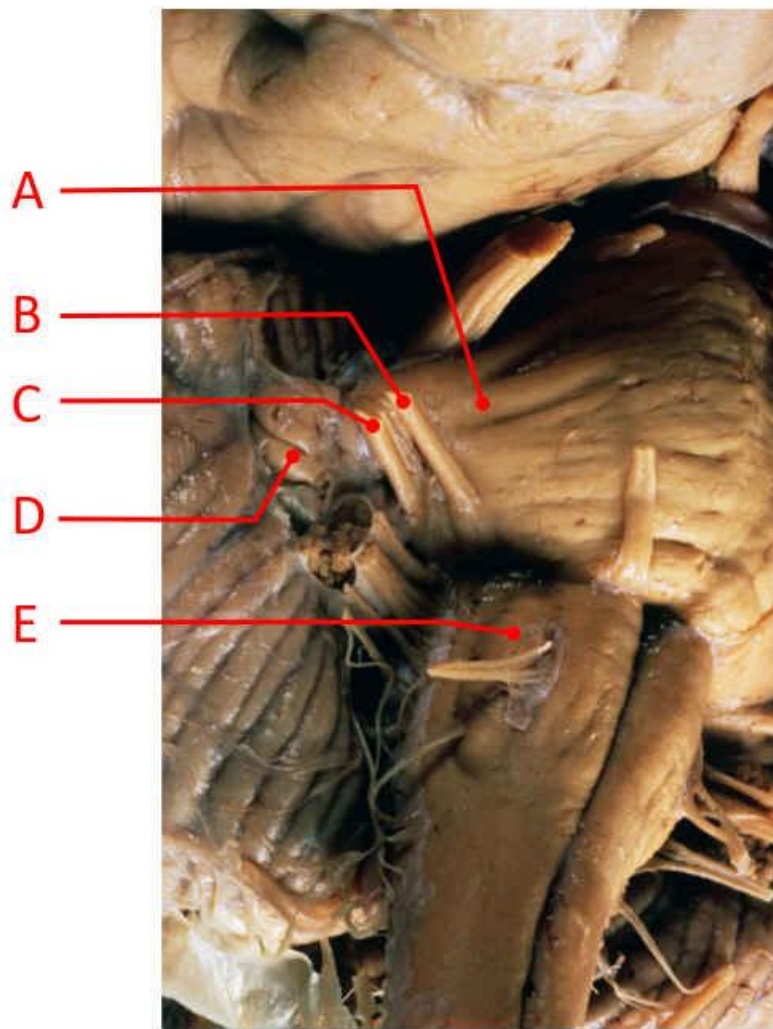
Question #: 26

A 44-year-old man comes to the physician because of a 4-month history of worsening right-sided hearing loss. Neurological examination shows right facial palsy and right sensorineural deafness. He also shows gait disturbance. Imaging studies show a lesion affecting the structure indicated by C in the attached figure. Which of the following outcomes is most likely?

- A. Persistent leftward deviation of the eyes
- B. Persistent rightward deviation of the eyes
- C. Constriction of the left pupil
- D. Constriction of the right pupil
- E. Dilation of the left pupil
- F. Dilation of the right pupil
- ✓G. Left nystagmus
- H. Right nystagmus

Rationale: The manifestations are consistent with a vestibular schwannoma. The clinical findings and the image are consistent with the lesion impacting the cranial nerves on the right. With damage to CN VIII on the right, the eyes would be expected to drift slowly to the right, with a fast saccade to the left. Hence, a left-beating nystagmus would be likely.

Attachment:



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Item Description: SOM.MK.II.BPM2.1.NB.4.NSCI.0253 Nystragmus

Question #: 27

A 1-year-old girl of immigrant parents is brought to the emergency department because of a tonic-clonic seizure. Her mother says the child's urine has an unusual odor. She had an uncomplicated vaginal delivery at 37-weeks gestation. She is exclusively formula fed and she has not been feeding well. Her weight and height are at 30th percentile, and head circumference is at 2nd percentile. Physical examination shows areas of skin and hair hypopigmentation. Neurological examination shows hypotonia in both upper and lower limbs. She is placed on a special phenylalanine-restricted diet with tyrosine supplementation. Several months later, on regular follow up she is found to have progressive neurologic deterioration. Laboratory findings show elevated prolactin. Which of the following supplementations is most likely prescribed to this patient?

- A. Vitamin B12
- B. Pyridoxine (vitamin B6)
- ✓C. Tetrahydrobiopterin
- D. Thiamine (vitamin B1)
- E. Pantothenic acid
- F. Folate (vitamin B9)

Rationale: The patient described in the vignette has phenylketonuria (PKU-II). Note the poor growth and microcephaly.

The urine odor is musty or mousey due to presence of phenylacetate, phenylpyruvate and phenyllactate in urine (phenylketones).

There is reduced melanin synthesis resulting in light colored skin.

The neurologic deterioration progresses even when placed on a phenylalanine-restricted diet. The prolactin level is elevated due to reduced formation of dopamine.

The patient has reduced synthesis of biopterin or reduced conversion of dihydrobiopterin to tetrahydrobiopterin (dihydrobiopterin reductase).

The patient would have reduced synthesis of catecholamines (low activity of tyrosine hydroxylase) and serotonin (low activity of tryptophan hydroxylase) leading to neurological features.

Item Description: PKU-II

Question #: 28

A 37-year-old man presents to the physician with a chief complaint of experiencing difficulty in chewing and biting food for the past four months. He has also observed a weight gain of 10 kilograms. His blood pressure measures at 150/94 mm Hg, while all other vital signs fall within normal ranges. Upon physical examination, the patient displays notable features such as frontal bossing, maxillary prominence, and a protruded mandible. The patient has undergone surgery to remove a pituitary tumor, but this procedure resulted in lesion to the posterior hypothalamus on both sides. What is the likely consequence of this hypothalamic lesion in this patient?

- ✓A. Hypothermia
- B. Obesity
- C. Disruption of the sleep–wake cycle
- D. Memory problem
- E. Dysregulation of water balance
- F. Hyperthermia

Rationale: Correct answer is A "hypothermia". Body temperature is regulated by opposing actions of the anterior and posterior areas of the hypothalamus. Posterior hypothalamus is responsible for

increasing body temperature and anterior hypothalamus is responsible for decreasing body temperature. Therefore, a lesion in the posterior hypothalamus will result in inability to regulate body temperature in a cold environment, leading to hypothermia. However, if fibers descending from the anterior hypothalamus are also damaged as they pass through the posterior hypothalamus, then a poikilothermic condition may arise.

Item Description: Hypothalamus and body temperature

Question #: 29

A 67-year-old comatose woman in the intensive care unit undergoes caloric testing. When cold water is injected into the left ear, both eyes turn to the left, and then slowly drift back into mid-position. When cold water is injected into the right ear, the right eye turns to the right, but the left eye stays in mid-position. After a while, the right eye drifts back into mid-position. The brainstem lesion is most likely located between which of the following structures?

- A. Left cerebral cortex / thalamus
- B. Right cerebral cortex / thalamus
- C. Left thalamus / midbrain
- D. Right thalamus / midbrain
- ✓E. Left midbrain / pons
- F. Right midbrain / pons
- G. Left pons / medulla
- H. Right pons / medulla

Rationale: Caloric testing applying cold water to the left and right ear induces eye movements, indicating intact afferent limbs of the vestibulo-ocular reflex bilaterally. Abduction of the left and right eye is normal, indicating intact abducens nuclei and nerves, bilaterally. Adduction of the left eye is impacted during lateral gaze to the right, indicating a left MLF lesion. Of the options given, a lesion between the left pons and left midbrain is most likely affecting the left MLF. Correct answer: E.

Item Description: NSCI.0252 Caloric testing and MLF lesion

Question #: 30

A 55-year-old man come to the physician because of subacute intensity of sound volume in his left ear in comparison to his right ear, for past 2 weeks. Diagnosis of left mandibular nerve schwannoma in the infratemporal fossa is made. Which structure will most likely be effected ?

- ✓A. Tensor tympani
- B. Stapedius
- C. Anterior auricular
- D. Posterior auricular

Rationale: The tensor tympanic branch from the mandibular nerve (CN V-3) innervates the tensor tympani muscle, which normally dampens vibrations of the malleus and tympanic membrane. The stapedius dampens vibrations of the stapes and is innervated by the stapedial branch of the facial nerve (CN VII). The 3 auricular muscles move the pinna and are each innervated by branches of the facial nerve.

Item Description: A 55-year-old man co